

Exploratory Data Analysis of Pharmaceutical Sale

Introduce the problem/goal:

In recent years, understanding the dynamics of pharmaceutical sales has become increasingly important for both businesses and public health officials. Different therapeutic drug classes (ATC categories) show distinct sales trends, and these patterns can reveal how consumer demand shifts with time, seasons, or other external influences.

In this project, I want to analyze six years of daily pharma sales data (2014–2019) and uncover the following:

- Which ATC categories dominate overall sales?
- How do sales trends evolve over time? Are there sharp increases or decreases in certain years?
- Do seasonal or weekday patterns exist that influence demand?
- Which categories move together (e.g., respiratory and allergy drugs)?

By studying these patterns, I hope to understand not only the market structure but also practical implications such as inventory planning, staff scheduling, and potential public health signals.

Introduce the Data:

The dataset I used for this project is from Kaggle: [Pharma Sales Data \(2014–2019\)](#). It contains six years of daily pharmaceutical sales, from January 2, 2014 through October 8, 2019, classified into eight ATC (Anatomical Therapeutic Chemical) categories.

Columns in the dataset include:

- datum – the date (string format, converted to datetime for analysis)
- year, month, hour, weekday_name – calendar helper columns for grouping and trend analysis
- m01ab – NSAIDs (acetic-acid derivatives, anti-inflammatory/analgesic)
- m01ae – NSAIDs (propionic-acid derivatives, ibuprofen-like)
- n02ba – Analgesics (salicylic-acid derivatives, aspirin-like)

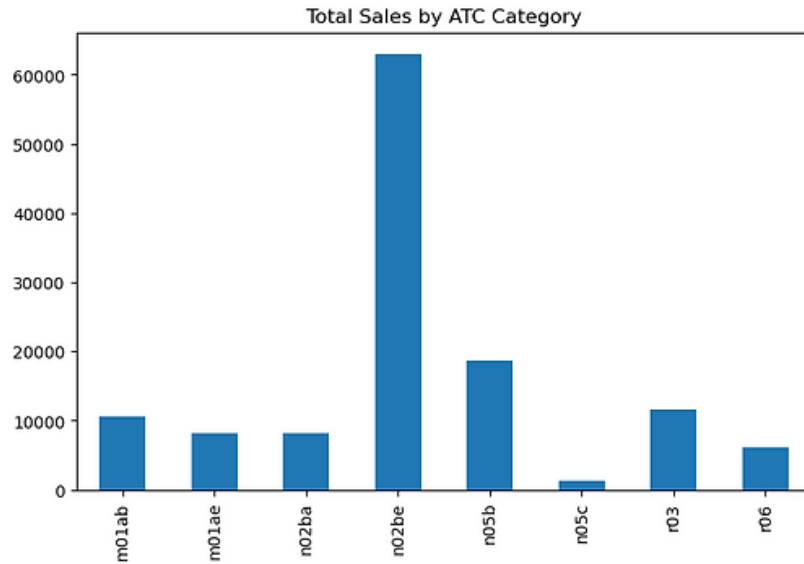
- n02be – Analgesic combinations (paracetamol combos, etc.)
- n05b – Anxiolytics
- n05c – Hypnotics and sedatives
- r03 – Respiratory drugs for obstructive airway disease (asthma, COPD)
- r06 – Antihistamines for systemic use

Pre-processing the Data:

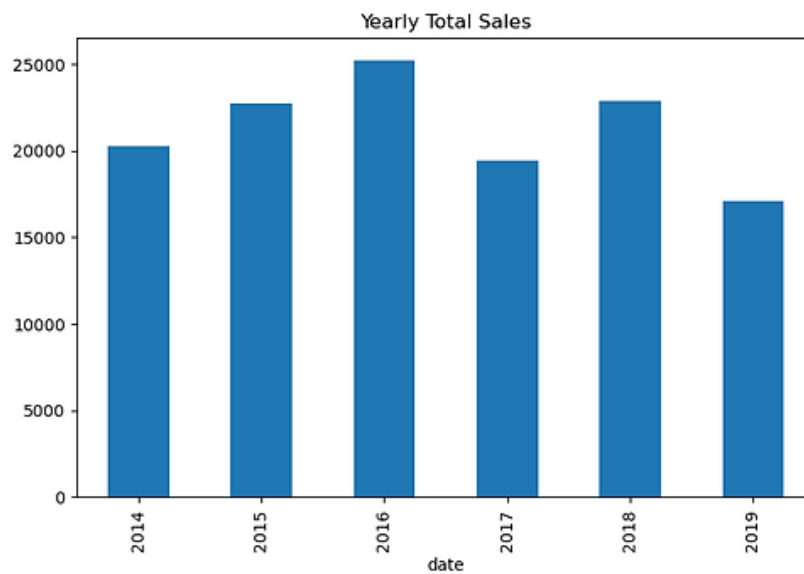
I began by cleaning the dataset to ensure it was consistent and analysis ready. First, I renamed all column names by converting them to lowercase and replacing spaces with underscores for consistency. Next, I parsed the datum column into a proper datetime column called date, which allowed me to group the data by months, years, and weekdays. I then checked the data types to make sure all the sales columns were numeric (float), ensuring that sums and averages would be calculated correctly. After that, I created a new column called total_sales by summing the eight ATC categories for each day, giving me a single metric to track overall trends. I also checked for missing values and duplicate dates none were found, and I kept zero values as they likely represented actual days with no recorded sales. To support trend analysis, I aggregated the data into monthly and yearly totals and computed year-over-year (YoY) percentage changes. Finally, I derived additional features such as month and weekday directly from the date column to cross-check the provided calendar helper columns. Together, these steps ensured the dataset was properly cleaned, consistent, and well-prepared for exploration and visualization.

Data Understanding/ Visualization:

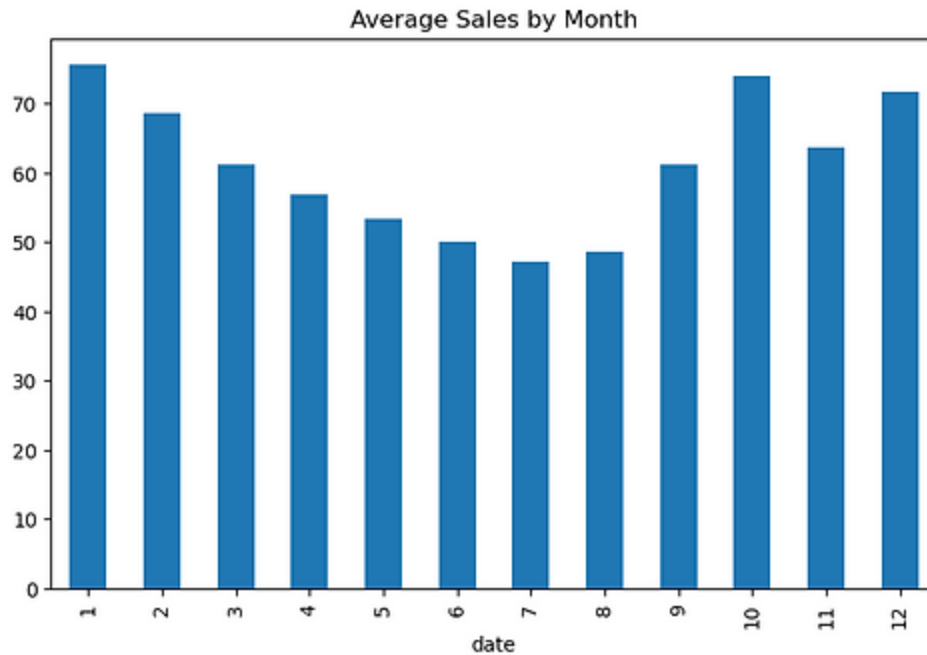
Total Sales by ATC Category (2014–2019): This chart shows the total sales for each ATC drug category over the six-year period. It highlights which categories dominate the overall pharmaceutical market and which ones play a smaller role.



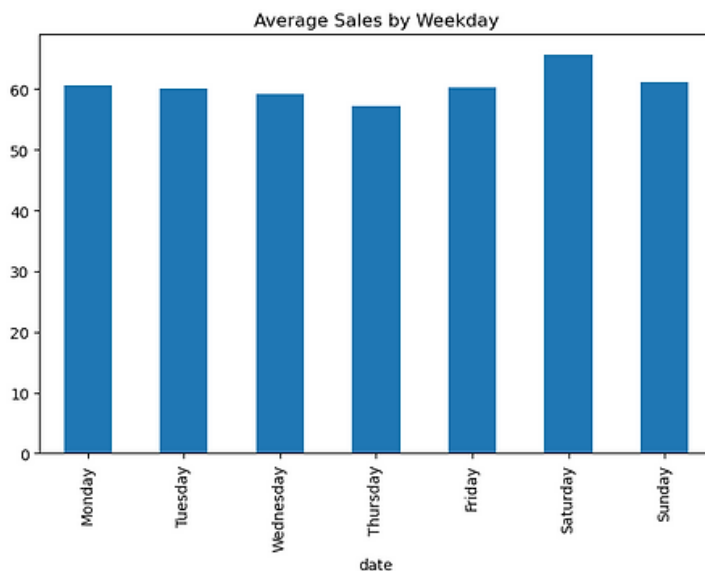
Yearly Total Sales (2014–2019): This bar chart shows the total pharmaceutical sales for each year in the dataset. It helps reveal overall growth and decline trends across the six-year period, including peak years and years of decline.



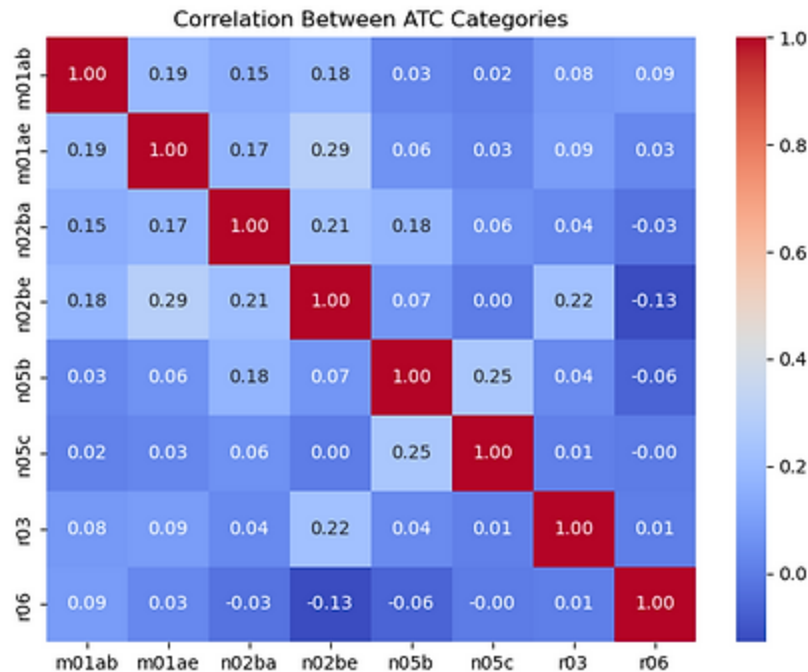
Average Sales by Month: This chart shows the average total sales for each calendar month across all years. It helps identify seasonal patterns, such as higher sales during the winter months and lower sales in the summer.



Average Sales by Weekday: This chart highlights the average sales for each day of the week. It shows whether sales activity tends to be stronger on weekdays or weekends.



Correlation Between ATC Categories: This heatmap shows the correlation between different ATC drug categories. Positive values indicate categories that move together, while negative or near-zero values suggest little to no relationship.



Storytelling:

Looking at the visualizations, a clear picture of the pharma sales data begins to emerge. The biggest driver of sales across these six years is N02BE (analgesic combinations), which makes up almost half of the total. Other categories such as N05B (anxiolytics) and R03 (respiratory drugs) also stand out, while smaller categories contribute much less. When I looked at the yearly totals, I noticed that sales climbed steadily through 2016, dipped in 2017, bounced back in 2018, and then dropped sharply in 2019. Part of that 2019 dip is simply because the dataset only goes up to October, but it still shows how sales can vary year to year. Seasonal patterns were also very clear: sales tend to spike in the winter months like January and December and fall off in the summer, which lines up with cold, flu, and allergy cycles. Even though weekdays show a slight pattern, Saturdays consistently had the highest sales, while Thursdays were the lowest. Finally, when I checked correlations, most categories seemed to move independently, meaning each class has its own demand cycle. Overall, these insights answered the questions I set out in the beginning about category mix, yearly trends, seasonality, weekday effects, and relationships between categories.

Impact:

The insights from this project could be very useful in practice. For example, knowing that sales pick up in the winter months and even a little more on weekends could help companies

plan with staffing and inventory. Focusing forecasting efforts on N02BE also makes sense, since it has such a big impact on overall totals. At the same time, I realize that visualizations like these could be misused. If someone only saw the dominance of pain-relief drugs, they might take it as encouragement to push more sales without considering whether that's appropriate for patients. The dataset itself has some limits too it doesn't include prices, locations, demographics, or promotional activity, and since 2019 is incomplete, we can't draw firm conclusions from that year. Also, without linking to health data like flu or allergy outbreaks, I can only guess the real reasons behind the seasonal spikes. Still, even with these limits, this project gave me a much clearer view of how pharmaceutical sales behave over time and showed me how data analysis can connect numbers to real-world patterns.

References:

<https://www.kaggle.com/datasets/milanzdravkovic/pharma-sales-data/data>